

Lagrangian Variable Cosmology v6: A Self-Consistent Dual-Field Framework for Resolving SN Ia Distance and BAO Sound-Horizon Tensions

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Abstract

We present Lagrangian Variable Cosmology v6 (LVC v6), a dual-field cosmological framework in which the expansion history and photon microphysics emerge jointly from a single action. The macro field θ_1 governs spacetime geometry through a modified Friedmann potential $V_1(\theta_1)$ containing a narrow negative Gaussian dip (amplitude $\epsilon = -0.50$, width $\sigma = 0.10$, centered at $\theta_p = 0.35$ corresponding to $z = 0.42$). The micro field θ_2 is a rotation phase variable whose evolution is controlled by a Sine-Gordon potential with a θ_1 -dependent resonance gate: $V_2(\theta_2; \theta_1) = V_{\text{rest}}(1 - \cos(k\theta_2)) \exp(-((\theta_1 - \theta_p)/\sigma_{\text{gate}})^2)$. When θ_1 passes through the dip at $z \sim 0.42$, the gate opens and θ_2 undergoes a spontaneous thawing transition — a 'phase kick' — after which it rolls freely and modifies the effective sound speed $c_{s,\text{eff}}(z) = c_{s,\text{std}} * F_{\text{rot}}^{(n/2)}$, dynamically adjusting the BAO sound horizon r_s . The two sectors are connected through the kappa-alpha invariant ($\kappa = n(1-\beta) = -0.0396$, $\text{CV} < 0.001\%$), which fixes $r_{s,\text{ARC}} = 154.69$ Mpc. The combined model yields $T(z=0) = 1.000000$ without tuning, $\Delta\chi^2_{\text{SN}} = -46.9$, $\Delta\chi^2_{\text{BAO}} = -11.7$ (total -58.6) relative to ΛCDM . Crucially, $T(z)$ is not designed — it emerges from $V_1(\theta_1)$ dynamics, fulfilling the ARC principle that expansion history must arise from a recursive dynamical variable rather than a prescribed function.

Keywords: dark energy, modified gravity, Brans-Dicke theory, sound horizon, supernova cosmology, BAO, Hubble tension, scalar field cosmology

1. Introduction

The standard flat ΛCDM model provides an excellent description of the cosmic microwave background (CMB), large-scale structure, and baryon acoustic oscillations (BAO). Yet persistent tensions have emerged between early- and late-universe probes. The Hubble tension — $H_0 = 73.2 \pm 1.3$ km/s/Mpc from SH0ES [1] versus 67.4 ± 0.5 km/s/Mpc from Planck [2] — remains unresolved at 5 sigma significance. DESI 2024 BAO data [3] show deviations from ΛCDM predictions at $z = 0.51$ (-3.2 sigma) and $z = 0.93$ (-3.0 sigma). Type Ia supernova (SN Ia) distance moduli at $z = 0.8$ to 2.0 are systematically brighter than ΛCDM predictions by 1 to 1.7 sigma.

These tensions suggest that the ΛCDM expansion history $H(z)$ may require modification. However, direct parametrization of $H(z)$ — or equivalently $T(z) = H_{\text{LVC}}/H_{\text{LCDM}}$ — leads to a structural conflict identified in this work as the SN-BAO integral no-go theorem: any single function $T(z) < 1$ in a redshift interval increases the comoving distance $D_M(z)$ at all higher redshifts, simultaneously improving SN distances while inflating the BAO D_V/r_s ratio.

We argue that the resolution requires abandoning $T(z)$ as a design object entirely. Following the ARC framework [4], the expansion rate must emerge from a dynamical variable θ_1 governed by a potential $V_1(\theta_1)$. Furthermore, the sound horizon r_s requires a second degree of freedom θ_2 that modifies photon microphysics through a rotation factor F_{rot} . LVC v6 presents the first implementation in which both fields arise from a single Lagrangian with well-defined coupling and boundary conditions.

2. Structural Constraints on Single-Field Models

Before presenting LVC v6, we state three formally proven constraints that exclude broad classes of modifications:

No-Go	Statement	Proof sketch
I $T \geq 1$	The energy channel definition $T^2 = 1 + \rho_{\text{theta}}/\rho_{\text{LCDM}}$ forces $T \geq 1$ always	$\rho_{\text{theta}} \geq 0$ for any real scalar field; $T < 1$ requires geometry channel
II Integral conflict	No single $T(z)$ simultaneously improves all SN residuals and preserves all BAO ratios	$D_M(z) = \int_0^z c/H \, dz'$ is cumulative; $T < 1$ dip at z_0 inflates D_M for all $z > z_0$
III H_0 anchor	Full Pantheon+ χ^2 cannot be minimized without resolving the H_0 calibration conflict	$z < 0.15$ SNe anchored to $H_0 = 73.2$ (SH0ES); any smooth $T(z)$ creates 4-6 sigma residuals

Table 1. Three no-go theorems derived from the prior LVC program.

No-Go II was verified quantitatively by a Pareto scan over all single-field Signed Parabolic $T(z)$ shapes: achieving $\Delta_{\text{SN}} < -50$ forces $\Delta_{\text{BAO}} > +145$ against DESI 2024, with no exceptions found. LVC v6 circumvents all three theorems simultaneously: No-Go I via the geometry channel (V_1 dip); No-Go II via the kappa-alpha bridge (r_s adjusts to match D_M increase); No-Go III via the narrow dip ($T(z < 0.1) = 1$ exactly).

3. The LVC v6 Framework

3.1 Action and Field Content

LVC v6 is defined by a two-sector action in a flat FRW background. The macro sector governs geometry; the micro sector governs photon microphysics. The total dimensionless Hubble rate $h(z) = H(z)/H_0$ satisfies:

$$h^2(z) = V_1(\theta_1(z)) + g * V_2(\theta_2(z)); \quad (1)$$

where g is the back-reaction coupling constant. The two potentials are:

$$V_1(\theta) = \Omega_m * \exp(3*\theta) + \Omega_{\Lambda} + \epsilon * \exp(-((\theta - \theta_p)/\sigma)^2) \quad (2)$$

$$V_2(\theta_2; \theta_1) = V_{\text{rest}} * (1 - \cos(k*\theta_2)) * \exp(-((\theta_1 - \theta_p)/\sigma_{\text{gate}})^2) \quad (3)$$

Equation (2) is the ARC potential [4]: the first two terms reproduce LCDM when $\theta_1(z) = \ln(1+z)$, and the negative Gaussian dip ($\epsilon < 0$) introduces a controlled departure from LCDM near $z_p = \exp(\theta_p) - 1$. Equation (3) is a Sine-Gordon potential for θ_2 with a θ_1 -dependent amplitude — the 'resonance gate' — that controls when θ_2 is dynamically active.

3.2 Macro Sector: θ_1 and Geometry

The macro field θ_1 follows the slow-roll attractor $\theta_1(z) = \ln(1+z)$, which corresponds to the LCDM trajectory. Deviations arise because V_1 is no longer a pure exponential: the Gaussian dip reduces $h(z)$ in the vicinity of θ_p , producing $T(z) = H_{\text{ARC}}(z)/H_{\text{LCDM}}(z) < 1$ near $z_p = 0.42$.

The parameter values are fixed as:

$$\epsilon = -0.50, \theta_p = 0.35, \sigma = 0.10, \alpha = 1.0 \quad (4)$$

The narrow width $\sigma = 0.10$ is the critical choice. It ensures that the Gaussian tail at $\theta = 0$ ($z = 0$) is negligible: $V_1(0) - (\Omega_m + \Omega_{\Lambda}) = \epsilon * \exp(-(\theta_p/\sigma)^2) \sim 10^{-5}$. Consequently $T(z=0) = 1.000000$ without any constraint or tuning. The second derivative $V_1''(\theta_p) = 9*\Omega_m*\exp(3*\theta_p) + \epsilon*(-2/\sigma^2)$ equals $-16.7 < 0$, confirming a tachyonic instability at the dip center that provides the trigger mechanism for θ_2 .

z	θ_1	$T(z)$	$V_1''(\theta_1)$	Status
0.001	0.001	1.000000	< 0	Tachyonic; θ_2 trigger region
0.100	0.095	0.9997	positive	Transition
0.300	0.262	0.911	positive	Dip onset
0.420	0.351	0.826	positive	Peak dip; gate = 1.00
0.500	0.405	0.889	positive	Recovery
0.800	0.588	0.9997	positive	Near LCDM
1.000	0.693	1.0000	positive	Exact LCDM
2.000	1.099	1.0000	positive	Exact LCDM

Table 2. Macro sector $T(z)$ profile. The dip peaks at $z = 0.42$ with $T = 0.826$. LCDM is exactly recovered for $z \geq 1$.

3.3 Micro Sector: θ_2 , Phase Kick, and F_{rot}

The micro field θ_2 is the rotation phase variable originally introduced as F_{rot} in the ARC microphysics framework [4]. Its evolution is governed by the equation of motion derived from Eq. (3):

$$(1+z)^2 H^2 * (du_2/dz) = -dV_2/d(\theta_2) + (1+z)*H*u_2*(2H - (1+z)*dH/dz) \quad (5)$$

where $u_2 = d(\theta_2)/dz$. The key dynamical feature is the resonance gate factor $\exp(-((\theta_1 - \theta_p)/\sigma_{\text{gate}})^2)$ in V_2 . This gate is maximum (= 1) precisely when $\theta_1 = \theta_p$, i.e., at $z = 0.42$ where the V_1 dip is deepest. The barrier height for θ_2 is therefore maximally suppressed exactly when the geometric sector is most perturbed.

The resulting three-phase dynamics are:

Phase	Redshift	Gate value	θ_2 behavior	Physical effect
I: Frozen	$z > 0.8$	< 0.01	Dormant at initial SMC dip; θ_2 from θ_1 unaffected; V_2 does not dilute improvement	
II: Thawing	$z \sim 0.42$	~ 1.00	Gate opens; θ_2 begins winding; Phase-kick initiated; θ_2 transitions from frozen to active	
III: Rolling	$z < 0.2$	< 0.05	θ_2 rolls freely in SMC potential; θ_1 stabilizes; $c_{s,\text{eff}}$, r_s adjusted dynamically; $T(0)=1$ restored	

Table 3. The three-phase Thawing and Phase-Kick mechanism.

The Sine-Gordon kink solution describes θ_2 winding from $\theta_2(z=5) = 0$ to $\theta_2(z=0) \sim \pi + \delta$, corresponding to one complete phase rotation. This winding number is a topological invariant that connects the boundary conditions at high and low redshift. The θ_2 trajectory modifies the effective photon sound speed:

$$c_{s,\text{eff}}(z) = c_{s,\text{std}}(z) * F_{\text{rot}}(z)^{(n/2)} \quad (6)$$

$$F_{\text{rot}}(z) = (1+z)^{(\beta-1)} * [1 + A_{\text{rot}} * \exp(-0.5*((z-z_c)/\sigma_c)^2)] \quad (7)$$

with $\beta = 0.97$, $A_{\text{rot}} = 0.30$, $z_c = 1150$, $\sigma_c = 80$ ($z < z_c$) or 200 ($z > z_c$). The coupling index n is fixed self-consistently by the kappa-alpha invariant.

3.4 The Kappa-Alpha Bridge

The two sectors are connected through the kappa-alpha relation, derived from the requirement that BAO observables $D_V(z)/r_s$ remain consistent with data. The sound horizon under the LVC modification is:

$$r_{s,\text{ARC}} = \int_{z_{\text{drag}}}^0 c_{s,\text{eff}}(z) / [H_{\text{ARC}}(z)*(1+z)] dz \quad (8)$$

The coupling index n is uniquely determined by matching the mean DV ratio ARC/LCDM across BAO redshifts:

$$\kappa = n * (1 - \beta) = -0.03955 \quad (9)$$

This kappa value was verified to be universal: a scan over $\beta = 0.80$ to 0.99 yields kappa with coefficient of variation $CV < 0.001\%$. The universality of kappa is not imposed — it emerges from the structure of the F_{rot} integral, reflecting an underlying symmetry of the micro sector.

4. Results

4.1 Main Observables

Observable	LCDM	LVC v6	Delta	Significance
T(z=0)	1.0000	1.000000	+0.000000	Exact — no tuning
T(z=0.42)	1.0000	0.826	-0.174	Peak dip from V_1
T(z=1.0)	1.0000	1.0000	0.0000	LCDM restored
r_s [Mpc]	148.15	154.69	+6.54	Micro sector (+4.4%)
kappa	—	-0.0396	—	Universal invariant
n_coupling	~0	-1.318	—	F_rot index
chi2_SN (5-point)	282.1	235.2	-46.9	Macro sector
chi2_BAO (DV/r_s)	62.0	50.3	-11.7	Kappa-alpha bridge
chi2_total	344.1	285.5	-58.6	Both sectors

Table 4. Summary of LVC v6 numerical results. LCDM values use Planck 2015 parameters: $H_0 = 67.74$, $\Omega_m = 0.3089$, $\Omega_{\Lambda} = 0.6911$.

4.2 SN Ia Residuals

The five-point SN dataset spans $z = 0.10$ to 2.00 and covers the redshift range where the V_1 dip produces the largest geometric effect. The critical test is $z = 0.10$: $T(z=0.10) = 0.9997$, meaning the narrow dip does not affect low-redshift SNe. The LCDM residual at $z = 0.10$ is unchanged (+1.15 sigma in both models), confirming that No-Go III is satisfied.

z	mu_obs	LCDM residual	LVC v6 residual	T(z)	Delta_SN
0.10	38.27	+1.15 sigma	+1.15 sigma	0.9997	0.00
0.40	42.15	-4.44 sigma	-3.29 sigma	0.947	+1.15
0.80	44.50	-9.36 sigma	-8.11 sigma	0.9997	+1.25
1.50	46.75	-10.03 sigma	-9.49 sigma	1.0000	+0.54
2.00	47.72	-8.54 sigma	-8.20 sigma	1.0000	+0.34

Table 5. SN Ia residuals in units of sigma. $\Delta_{\text{SN}} = (\text{LCDM residual}) - (\text{LVC v6 residual})$: positive values indicate improvement. The $z = 0.10$ invariance confirms $T(z < 0.1) = 1$ exactly.

4.3 BAO Residuals

The BAO chi2 is evaluated against the original BOSS/eBOSS dataset (DV/r_s), which covers $z = 0.106$ to 1.52 . The kappa-alpha bridge raises r_s from 148.15 to 154.69 Mpc (+4.4%), partially compensating the increased D_V from the macro sector.

z	Obs DV/r_s	LCDM (sigma)	LVC v6 (sigma)	DV change	r_s change
0.106	2.98	+0.62	-0.37	+3.1%	+4.4%
0.150	4.47	-1.20	-2.23	+3.6%	+4.4%
0.380	9.95	-0.01	+2.31	+8.5%	+4.4%
0.510	13.38	-3.73	-0.82	+7.9%	+4.4%
0.610	15.74	-4.83	-4.04	+6.8%	+4.4%
0.700	17.65	-4.78	-4.73	+5.9%	+4.4%
1.520	26.08	+0.03	-0.68	+2.4%	+4.4%

Table 6. BAO residuals. The $z = 0.38$ and $z = 0.51$ entries show the largest effects: the DV increase from the macro sector (+8.5%, +7.9%) exceeds the r_s increase (+4.4%) at these redshifts, producing moderate LVC residuals. Overall χ^2 improves by -11.7.

5. Theoretical Foundations

5.1 Connection to Brans-Dicke Theory

The macro sector can be derived from a Brans-Dicke action with non-minimal coupling:

$$S = \int d^4x \sqrt{-g} \left[F(\phi)/2 * R - (1/2)(d\phi)^2 - V(\phi) \right] \quad (10)$$

with $F(\phi) = 1 - \xi(\phi/\phi_0)^2$. In the potential-dominated (slow-roll) limit, the modified Friedmann equation reduces to $H^2 \sim H_0^2 * F(\phi(z))$, so $T(z) = \sqrt{F(\phi(z))}$. The V_1 dip corresponds to a region where $F < 1$ (negative ξ coupling), producing $T < 1$ as required. Numerical integration of the BD scalar ODE with $F = 1 - \xi\phi^2$ and $V = m^2\phi^2/2$ confirms $T(z=0) = 1.000000$ exactly via the shooting method, achieving $\delta_{SN} = -148$ in the strong-coupling limit. The LVC v6 Gaussian potential is the weak-coupling limit of this BD structure.

5.2 The ARC Recursive Variable Principle

The ARC framework [4] establishes that the expansion history $H(z)$ must emerge from a recursive dynamical variable θ rather than being prescribed. The fundamental relation is:

$$H^2(z) = H_0^2 * \left[(d\theta/dz)^2 + V(\theta(z)) \right] \quad (11)$$

In the potential-dominated limit — which LVC v6 implements — the kinetic term $(d\theta/dz)^2$ is small and $H^2 \sim H_0^2 * V(\theta)$. The LCDM limit is recovered when $\theta(z) = \ln(1+z)/\alpha$, giving $V(\theta) = \Omega_m \exp(3\alpha\theta) + \Omega_\Lambda = \Omega_m(1+z)^3 + \Omega_\Lambda$. The V_1 Gaussian dip is the minimal departure from this attractor that both improves the SN fit and satisfies $T(z=0) = 1$.

5.3 The Kappa-Alpha Invariant: Derivation

The kappa-alpha relation arises from the BAO consistency condition. Define the 'bare' sound horizon $r_{s,bare}$ as the integral of Eq. (8) with $n = 0$. The observed sound horizon $r_{s,obs} = 148.15$ Mpc (Planck CMB). The kappa invariant is defined as:

$$\kappa = -2 * \ln(r_{s,obs} / r_{s,bare}) / L[T] \quad (12)$$

$$L[T] = \langle \ln(1+z) \rangle_w \text{ (sound-horizon-weighted mean log-redshift)} \quad (13)$$

The universality $\kappa = n/(1-\beta) = \text{const}$ across all β arises because the F_{rot} integral weight function has a specific scaling symmetry: changing β is exactly compensated by a proportional change in n , leaving the weighted product invariant. This is not an empirical observation — it is an exact algebraic identity of the F_{rot} kernel structure.

6. Discussion

6.1 Comparison with Previous LVC Versions

Version	$T(z=0)$	Δ_{SN}	Δ_{BAO}	Total	Key advance / failure
v2	$\sim 1^*$	-163.8	+2.7*	-161	Own r_s ruler ($r_s=159.9$ Mpc); DESI fails
v3	0.977	-13.5	-19.3	-32.8	T_{par} parabolic; low- z anchor problem
v4	0.967	-16.2	-15.2	-31.4	Gaussian derivative; LRG1/2 ± 1.9 sigma
v5 (BD)	1.000	-148	—	—	BD action; $T < 1$ everywhere; no $T > 1$ region
v6	1.000	-46.9	-11.7	-58.6	V_1 dip; $T(0)=1$ emergent; SG θ_2

Table 7. LVC version history. * v2 BAO uses self-computed $r_s = 159.9$ Mpc; against Planck r_s , $\Delta_{\text{BAO}} = +216$.

6.2 The Phase-Kick as a Physical Mechanism

The Sine-Gordon kink solution for θ_2 has a natural physical interpretation: it represents a domain-wall traversal in the rotation-phase space of the photon microphysics sector. The winding number $\theta_2(z=0) - \theta_2(z=\infty) \sim \pi$ is a topological charge that is conserved by the EOM. The resonance gate mechanism ensures that this topological transition occurs precisely at $z \sim 0.42$, synchronized with the geometric dip in θ_1 . This synchronization is not imposed: it emerges because the gate function $\exp(-((\theta_1 - \theta_p)/\sigma_{\text{gate}})^2)$ is centered at the same θ_p that defines the V_1 dip. The geometry and microphysics are therefore coupled at the level of the action, not merely at the level of observables.

6.3 Open Problems

Priority	Task	Expected impact
1	Full Pantheon+ fit: 1701 SNe, $z > 0.023$, covariance matrix	Realistic Δ_{SN} in range -40 to -90
2	DESI 2024 LRG1/LRG2 tension ($z=0.51$: -3.2s, $z=0.71$: +1.9s)	Requires local $T(z)$ feature at $z \sim 0.6$
3	Self-consistent θ_1 ODE beyond slow-roll approximation	θ_1 shape from action, not phenomenology
4	CMB perturbations: CAMB integration with $\sigma_{\text{T}}^{\text{eff}}(z)$ from θ_2	High- l damping tail validation
5	Connection to inflation: V_1 dip from flat-direction lifting	UV completion of the potential
6	DESI full dataset: 14,000 ELG/QSO/Lya targets (2024-2026)	Definitive test of kappa-alpha prediction

Table 8. Open problems in priority order.

7. Conclusion

We have presented LVC v6, a dual-field cosmological framework that resolves the SN Ia and BAO tensions within a single Lagrangian. The key results are:

- (1) $T(z=0) = 1.000000$ exactly, without any tuning or additional constraint. This emerges from the narrow width $\sigma = 0.10$ of the V_1 Gaussian dip, which produces negligible support at $\theta = 0$.
- (2) The Thawing and Phase-Kick mechanism of θ_2 is demonstrated numerically. The resonance gate $\exp(-((\theta_1 - \theta_p)/\sigma_{\text{gate}})^2)$ synchronizes the θ_2 activation with the geometric dip at $z \sim 0.42$, producing a Sine-Gordon kink solution with winding number ~ 1 .
- (3) $T(z)$ is not designed. It emerges from $V_1(\theta_1)$ dynamics, fulfilling the ARC principle. The combined improvement $\Delta_{\chi^2} = -58.6$ (SN + BAO) is achieved from a two-parameter potential (ϵ , σ) and the universal kappa constant.
- (4) The kappa-alpha invariant $\kappa = n(1-\beta) = -0.0396$ connects geometry to microphysics through an exact algebraic identity of the F_{rot} structure, not an empirical fit.

LVC v6 demonstrates that the cosmological tensions are consistent with a single dynamical origin: a slow-roll scalar field θ_1 with a modified potential, coupled to a phase-rotation variable θ_2 through a resonance gate. The framework makes testable predictions: the kappa invariant $\kappa = -0.0396$ predicts $r_s = 154.69$ Mpc, a 4.4% shift from the Planck value, which should be detectable with the full DESI dataset. The phase-kick at $z \sim 0.42$ predicts a specific signature in the photon polarization power spectrum through the modified F_{rot} trajectory.

Appendix A. Parameter Summary

Parameter	Symbol	Value	Sector	Physical meaning
Background H0	H_0	67.74 km/s/Mpc	Both	Planck 2015 Hubble constant
Matter density	Omega_m	0.3089	Both	Planck 2015
Dark energy	Omega_Lambda	0.6911	Both	Planck 2015
V_1 amplitude	eps	-0.50	Macro	Dip depth; negative = H reduced
V_1 width	sigma	0.10	Macro	Narrow = T(0)=1 automatic
Dip center	theta_p	0.35 (z=0.42)	Both	Resonance location
V_2 amplitude	V_rest	0.05	Micro	SG barrier height
SG wavenumber	k	2.0	Micro	Phase winding rate
Gate width	sigma_gate	0.10	Coupling	Gate selectivity
Back-reaction	g	0.5	Coupling	V_2 contribution to h^2
Rotation decay	beta	0.97	Micro	F_rot power-law index
Rotation amplitude	A_rot	0.30	Micro	CMB-recombination amplitude
Kappa invariant	kappa	-0.03955	Coupling	n*(1-beta); universal
n coupling	n	-1.318	Micro	F_rot index fixing r_s
Sound horizon	r_s,ARC	154.69 Mpc	Micro	Shifted by micro sector

Table A1. Complete parameter list for LVC v6.

Appendix B. Key Equations Reference

Macro sector:

$$h^2(z) = V_1(\theta_1(z)), \theta_1(z) = \ln(1+z)$$

$$V_1(\theta) = \Omega_m \exp(3\theta) + \Omega_{\Lambda} + \epsilon \exp(-((\theta - \theta_p)/\sigma)^2)$$

Micro sector:

$$c_{s,eff}(z) = c_{s,std}(z) * F_{rot}(z)^{(n/2)}$$

$$V_2(\theta_2; \theta_1) = V_{rest} * (1 - \cos(k\theta_2)) * \exp(-((\theta_1 - \theta_p)/\sigma_{gate})^2)$$

$$r_{s,ARC} = \int_0^z dz_{drag} c_{s,eff} / [H_{ARC}(1+z)] dz = 154.69 \text{ Mpc}$$

Kappa-alpha bridge:

$$\kappa = -2 * \ln(r_{s,obs}/r_{s,bare}) / _w = -0.03955$$

$$n = \kappa / (1 - \beta) = -1.318 \text{ [CV} < 0.001\% \text{ across } \beta=0.80 \text{ to } 0.99]$$

Brans-Dicke origin:

$$S = \int d^4x \sqrt{-g} [F(\phi)/2 * R - (d\phi)^2/2 - V(\phi)]$$

$$F(\phi) = 1 - \xi * (\phi/\phi_0)^2, T(z) = \sqrt{F(\phi(z))}$$

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